

Lecture 36 — Characteristic equation, 2×2 eigenproblems, and complex numbers

Wed Nov 18

Last updated: October 8, 2026.

Reading: [Strang §6.1](#), characteristic equation and the complex-eigenvalue (90° rotation) example. Related objectives: [Lectures](#). [Schedule](#).

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, students should be able to:

1. Interpret $a + bi$ algebraically and as a point/vector in the complex plane.
2. Perform the basic complex arithmetic needed for eigenvalue calculations.
3. Form and solve $\det(A - \lambda I) = 0$ for a 2×2 matrix.
4. Find corresponding eigenvectors.
5. Check real or complex eigenpairs by substitution.

Warm-up

Monday's rotation $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ had no real eigenvector.

- (a) Form $R - \lambda I$ and compute $\det(R - \lambda I)$ as a quadratic in λ .
- (b) What are the roots? Are they real?

Fill in

Complex numbers. A complex number is $z = a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = \underline{\hspace{1cm}}$. Plot z as the point (a, b) in the complex plane.

Arithmetic (what we need):

- Add/subtract: combine real and imaginary parts.
- Multiply: $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$.
- Conjugate: $a + bi = \underline{\hspace{1cm}}$.
- Magnitude: $|a + bi| = \sqrt{a^2 + b^2}$.

Characteristic equation. Eigenvalues of square A solve

$$\det(A - \lambda I) = 0.$$

For 2×2 , this is a equation in λ .

Eigenvectors. For each eigenvalue λ , solve

$$(A - \lambda I)v = 0, \quad v \neq 0$$

(null space of $A - \lambda I$). Scale freely — any nonzero multiple is fine.

Check. Always verify $Av = \lambda v$ by substitution (works for real or complex pairs).

Worked in class

W1. Complex drill. Let $z = 2 + i$ and $w = 1 - 3i$.

- (a) Compute $z + w$, zw , and $\bar{z}$.
- (b) Plot z and $\bar{z}$ in the complex plane.

W2. Real 2×2 by hand. Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

- (a) Expand $\det(A - \lambda I)$ and solve for λ .
- (b) For each eigenvalue, find one eigenvector.
- (c) Check one pair by computing Av and λv .

W3. Rotation eigenvalues. For $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$:

- (a) Confirm the characteristic polynomial is $\lambda^2 + 1 = 0$, so $\lambda = \pm i$.
- (b) For $\lambda = i$, solve $(R - iI)v = 0$ and give one complex eigenvector.
- (c) Check $Rv = iv$ by matrix-vector multiplication (complex arithmetic).

W4. Interpretation preview. Complex conjugate eigenvalues of a real matrix often signal _____ (rotation / oscillation) in the real plane — Friday's lab makes this concrete with software.

Your turn

Y1. Compute $(3 - 2i)(1 + i)$ and $|3 - 2i|$.

Y2. For $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$, form $\det(A - \lambda I)$, find both eigenvalues, and an eigenvector for $\lambda = 3$.

Y3. Check: with $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $v = (1, -i)$, verify $Av = iv$ (or explain the correct λ if this v is for $\lambda = -i$).

Y4. True or false: every real 2×2 matrix has two real eigenvalues.

Answers (Your turn)

Y1. $(3 - 2i)(1 + i) = 3 + 3i - 2i - 2i^2 = 5 + i$. Magnitude: $|3 - 2i| = \sqrt{9 + 4} = \sqrt{13}$.

Y2. $A - \lambda I = \begin{bmatrix} 1 - \lambda & 2 \\ 0 & 3 - \lambda \end{bmatrix}$, so $\det = (1 - \lambda)(3 - \lambda)$. Eigenvalues $\lambda = 1$ and $\lambda = 3$. For $\lambda = 3$: $(A - 3I)v = \begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix} v = 0$ gives $v_1 = v_2$, e.g. $v = (1, 1)$.

Y3. $Av = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} i \\ 1 \end{bmatrix}$ and $iv = \begin{bmatrix} i \\ 1 \end{bmatrix}$. Yes — $Av = iv$ (so this v matches $\lambda = i$).

Y4. **False.** The characteristic polynomial may have complex roots (e.g. 90° rotation: $\lambda = \pm i$).